

HPCL Green

Data Collection & Inverter Mapping — Site Status Report

(27-07-2026)

Plant Overview

Parameter	Details
Site	HPCL Green Solar Plant
Total Capacity	Not yet provided (pending from Plant engineer)
Panel Count	~293 panels (approx.) — updated count expected on next site visit
Panel Breakdown	45 panels of 540W; remaining panels are 320W (approx.)
Inverters	3 inverters — data to be extracted from all three

Inverter & MFM details

- Total Inverter Count** : 3
- Inverter Make & Model** : EVVO 33000TLG23P
- Inverter Type** : EVVO Grid-Tied Solar Inverter (Three-Phase)
- Individual Power Capacity** : 33 kW (33,000 W)
- Max output Power/ Current** : 36.3 kVA , 3 x 53 A
- Total MFM Count** : 1
- MFM Make & Model** : Secure Elite 440

The EVVO 33000TLG23P is an outdoor-rated inverter featuring an IP65 ingress protection rating. On the AC grid side, it delivers a nominal output power of 33,000 W (33 kW) and a maximum output power of 36,300 VA (36.3 kVA), supporting a maximum output current of 3 x 53 A alongside a nominal grid voltage of 400 Vac (3-Phase / Neutral / Protective Earth) and a power factor greater than 0.99. For its DC solar input capacity, the unit accommodates a maximum DC input voltage of 1,100 V and operates within an MPPT voltage range of 230 V to 960 V, while handling a maximum input current of 30 A / 36 A and a maximum short circuit current (Isc) of 37.5 A / 37.5 A.

Inverter Com & Polling Status

Equipment	Operational Status	RS485 Data Capture Notes
Inverter 1	Modbus Unreachable	Readings are not captured (getting Inverter 2, Inverter 3 data). Due to miss match in inverter ID missing Inverter 1 data
Inverter 2	Online	Readings are captured (getting Inverter 2, Inverter 3 data).
Inverter 3	Online	Readings are captured (only Inverter 3 reading is captured).

Inverter Connectivity

Current Understanding:

- Per HPCL, Inverter 3 is expected to provide combined data for Inverters 1 and 2, as the three are connected in a loop via Rs485.
- In practice, Inverter 3 is not returning data for the other two inverters as expected.
- Inverters 1 and 2 can each be read individually, and each returns data for two inverters (2, 3 inverters only).
- The identity of the "other" inverter paired with each reading still needs to be confirmed to fully map the loop.

Sample readings have been collected from each inverter (a 36-register response is generated by the inverters).

Attached Data Files in Git

- `Evvo_rs485_Out_SniffText.txt`
- `evvo_raw_Invertor1.jsonl`, `evvo_raw_Invertor2.jsonl`, `evvo_raw_Invertor3.jsonl`
- `All3-inverter-combined_decode.excel`

Wattmon Master & Polling Data

It has been observed that a Watmon MINI3-4G is connected to Inverter 3, which acts as the Master polling for each Inverter. Captured some of the polling data from this Modbus master (Watmon device) to inverters.

Raw Polling Data Sample:

```
03 06 90 00 00 64 a4 c3 03 06 90 00 00 64 a4 c3 0b 03 00 63 00 16 34 b0 0c 03 00 63 00 16 35 07
```

Decoded Sample:

03 06 90 00 00 64 a4 c3 → slave 3, WRITE reg 0x9000 = 100 (Wattmon command)

0b 03 00 63 00 16 34 b0 → slave 11, READ 22 regs from addr 99 (meter)

0c 03 00 63 00 16 35 07 → slave 12, READ 22 regs from addr 99 (meter)

Site Photographs

Inverter 1



Inverter 2



Inverter 3



Some images of Inverters and connections:



Hardware Connections & Wiring



Decoder Values from Registers

Inverters

- **Inverter 1:** [**SILENT**] polls=43 resp=0 → (not generating / offline)

Issue Identified: Inverter 1 is mapped with Id : 00 but Master polling using 01

- **Inverter 2:** [**NORMAL**] 0s ago | polls=78 resp=54

Yield: Today 57.66 kWh | Total 80992 kWh | Power 0.75 kW

AC: R 240.0V 1.4A | Y 236.7V 1.3A | B 238.6V 1.3A | 50.07Hz

DC: PV1 582.8V 0.7A | PV2 509.7V 0.6A | Temp 60C

- **Inverter 3:** [**NORMAL**] 0s ago | polls=67 resp=43

Yield: Today 88.66 kWh | Total 74403 kWh | Power 1.25 kW

AC: R 238.2V 1.9A | Y 235.9V 1.9A | B 237.9V 1.9A | 50.07Hz

DC: PV1 581.9V 1.5A | PV2 569.1V 0.8A | Temp 60C

Meters (MFM)

- **Meter 11:** polls=81 resp=0 → SILENT
- **Meter 12:** polls=72 resp=0 → SILENT
- **Meter 13:** polls=15207 resp=0 → SILENT

Issue: Need to check if MFM reading is generated via polling

Weather Monitoring System (WMS)

WMS is not installed at this location. Other sites without WMS have been flagged for installation.

Identified Issues

1. Inverter 1 is mapped with Id: 00 but Master polling using 01
2. MFM data is not generated. need to check where we have this MFM data.
3. Missing WMS in the Plant to calculate power efficiency
4. String wise Power calculations is not generated from the Inverters (need to check if inverter can generate those)
5. Wattmon credentials are not provided from the plant to config it

Next Steps

- Collect data from Inverter 1 Need to fix Either Polling Id or Updated ID in the Inverter
- Tap all the Wattmon Master polling commands.
- Capture MFM reading if data is created.
- Analyze the inverter connection loop — likely requires a physical site visit to confirm wiring/connections.
- Status as of site visit — panel count and capacity figures pending confirmation on next visit.
- Map collected inverter readings against the API endpoint response data to check consistency.
- Complete WMS installation analysis for this and other flagged sites.